

ANGUS BARLOW

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SOFTWARE ENGINEERING | LLM PIPELINES & EVIDENCE SYNTHESIS

PROFESSIONAL SUMMARY

Software engineer who builds evaluation infrastructure and treats measurement as the product. My llm-posttraining repo runs staged behavioral evals with committed raw generations; my oncology-coscientist binds every reported number to recorded computation via a deterministic verifier; my comp-tox pipeline reports out-of-distribution metrics as findings.

TECHNICAL SKILLS

Evaluation: eval harnesses, benchmark construction, frozen splits, behavioral test sets, per-regime metric reporting, committed raw outputs

Engineering: Python, FastAPI, pytest (45+ test suites), CI-friendly repos, Docker, Snakemake, SQL

ML systems: PyTorch, JAX/Flax, SFT/DPO/GRPO post-training, diffusion, provider-API integration patterns

Practice: failure-mode documentation, provenance manifests, hash-chained audit logs, deterministic replay

EXPERIENCE

KBI Biopharma | Applied AI Intern | 2026

- Built a Python/FastAPI prototype for AI-assisted review of biopharmaceutical deviation investigations, integrating heterogeneous records, typed schemas, retrieval/cross-referencing, analytics, and traceable outputs.
- Stood up an evaluation harness that scores review outputs against documented edge cases before they reach stakeholders.
- Built enterprise-data acquisition automation over Microsoft Graph and SharePoint with pagination, state recovery, failure logging, and diagnostics.
- Designed a Bedrock/Claude integration architecture where model authority is bounded by recorded evidence, matching QMS audit expectations.

SELECTED TECHNICAL PROJECTS

LLM Post-Training | SFT, DPO Failure Diagnosis, GRPO Repair

Python | PyTorch | SFT/DPO/GRPO | SmolLM2-135M | github.com/barlowa124/llm-posttraining

- Hand-rolled DPO collapsed deployed behavior at 100% train accuracy; diagnosed the zero-gradient reward structure; shaped GRPO repaired the collapse with replications across rollout seeds.
- GRPO from a healthy policy is provably inert (60/60 no-signal steps, bit-identical output), the negative control other writeups skip.
- All six checkpoints and raw per-stage generations published on Hugging Face, including the collapsed DPO one.

Oncology Co-Scientist | Survival ML with a Deterministic Verifier

Python | LangGraph | Cox PH | Random Survival Forest | TCGA | github.com/barlowa124/oncology-coscientist

- Cohort-configurable survival pipeline (LUAD/GBM/BRCA) with frozen splits and explicit abstention; LangGraph agents draft reports while a deterministic verifier binds every number to recorded computation.
- The verifier caught fabricated cohort sizes, invented metrics for abstained models, and misattributed model blocks; rejected drafts and transcripts preserved; 45-test suite.

Comp-Tox Pipeline | Tox21 Evaluation with Calibration + AD

Python | PyTorch + JAX/Flax | scaffold-split LR/RF/GIN | github.com/barlowa124/comp-tox-pipeline

- Reproducible Tox21 pipeline: scaffold-split logistic/RF/graph models with calibration, conformal coverage, and applicability-domain conditioning per endpoint.
- JAX/Flax port verified for numerical parity with PyTorch; real multi-process DDP.

- The NR-ER applicability-domain inversion only surfaced because out-of-domain metrics were measured and documented as a finding.

Vector-DB MCP | Local Vector Database over MCP

Python | MCP/stdio | IVF + HNSW | Pinecone API surface | github.com/barlowa124/vector-db-mcp

- Local vector DB exposing Pinecone's API over MCP: exact flat, IVF, HNSW scans with a committed benchmark where flat exact wins at scale and two construction bugs were caught by recall measurement.

OPEN SOURCE CONTRIBUTIONS

chaidiscovery/chai-lab (open PR) | #431 adds a single-chain aggregate confidence score for monomer predictions.

scverse/scanpy (open PRs) | #4383 and #4385 fix user-supplied hue handling and non-string multi-column groupby crashes across shared plotting code.

OpenADMET/openadmet-models (open PRs) | #607-#610: eval-split scaler leakage fix, calibration soundness, scaffold applicability domain (Tanimoto/Morgan), configurable n_jobs.

ADDITIONAL PROJECTS

Bioprocess Decision Runtime | Typed re-execution of Gemma 3 internals with bit-exact conformance checks and hash-chained audit logs.

Protein Diffusion | Conditional DDPM on a measured fitness oracle; memorization reported as a headline metric.

EDUCATION

North Carolina State University | Master of Biomanufacturing (In Progress) | Raleigh, NC

Southern New Hampshire University | B.S. Computer Science; B.S. Environmental Science | Jan 2026

Magna Cum Laude | GPA 3.702 | CS coursework: Data Structures & Algorithms, Software Security, Software Testing/Automation QA, Client/Server Development